FACULTY OF ECONOMICS
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Microeconometrics | Chapter 7

Regression Discontinuity Designs

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Introduction

- Regression Discontinuity (RD) is a powerful causal inference method, particularly useful when randomized controlled trials are not feasible.

- RD designs exploit "natural experiments," where treatment assignment depends on an observed score crossing a predetermined cutoff, creating comparable groups immediately above and below this threshold

A guide to regression discontinuity designs in medical applications" by Cattaneo, Keele, and Titiunik (2023)

- The paper outlines two main conceptual frameworks for analyzing RD designs:
  - Continuity-based framework: This method assumes continuous potential outcomes at the cutoff. It calculates the average treatment effect specifically at the cutoff point, known as the sharp RD treatment effect.
  - Local randomization framework: This framework assumes local random assignment around the cutoff, allowing analysis akin to randomized experiments within a small "window" near the threshold

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- Diagnostic and Falsification Tests:
  - The paper stresses the importance of validating RD assumptions through diagnostic and falsification tests, including:
    - Checking the distribution of scores around the cutoff to detect possible manipulation or sorting.
    - Conducting balance tests on predetermined covariates to ensure comparability of groups at the threshold.
    - Implementing weak IV tests for fuzzy RD designs to confirm the strength of the instrumental variable (i.e., the score relative to treatment uptake).

- Empirical Applications:
  - The paper illustrates these concepts through several empirical examples:
    - ART (anti-retroviral therapy) in HIV patients in South Africa, examining the effect of immediate versus deferred ART based on CD4 count thresholds.
    - Chemotherapy guidelines for breast cancer recurrence based on genetic test scores.
    - Patient cost-sharing policies and their impact on healthcare utilization in Taiwan

- Types of RD Designs:
  - Sharp RD designs assume perfect compliance with treatment assignment rules. Units above the cutoff always receive the treatment, while those below do not.
  - Fuzzy RD designs acknowledge imperfect compliance, common in biomedical contexts. Here, crossing the cutoff strongly influences—but does not perfectly determine—treatment assignment, requiring instrumental variables (IV) techniques to estimate effects accurately.

- Estimation and Inference Methods:
  - The authors emphasize modern, robust methods for estimation and inference, including:
    - Optimal bandwidth selection (determining how close to the cutoff observations must be included).
    - Local polynomial regression techniques for estimating treatment effects.
    - Robust bias-corrected inference methods to enhance accuracy and reliability of RD estimates

Practical Recommendations:

- Always start with graphical analysis (e.g., RD plots) for preliminary understanding.
- Localize estimation and inference around the cutoff, avoiding global methods that can introduce bias.
- Use continuity-based methods for continuous scores and local randomization methods for discrete scores.
- Perform rigorous falsification and validation tests to ensure reliability and robustness of the RD results

All codes and data are in this link

REPLICATION · RD Packages

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